Simulating the Floor: An Archive-Grounded Generative Agent Sandbox of Dutch Parliamentary Debate

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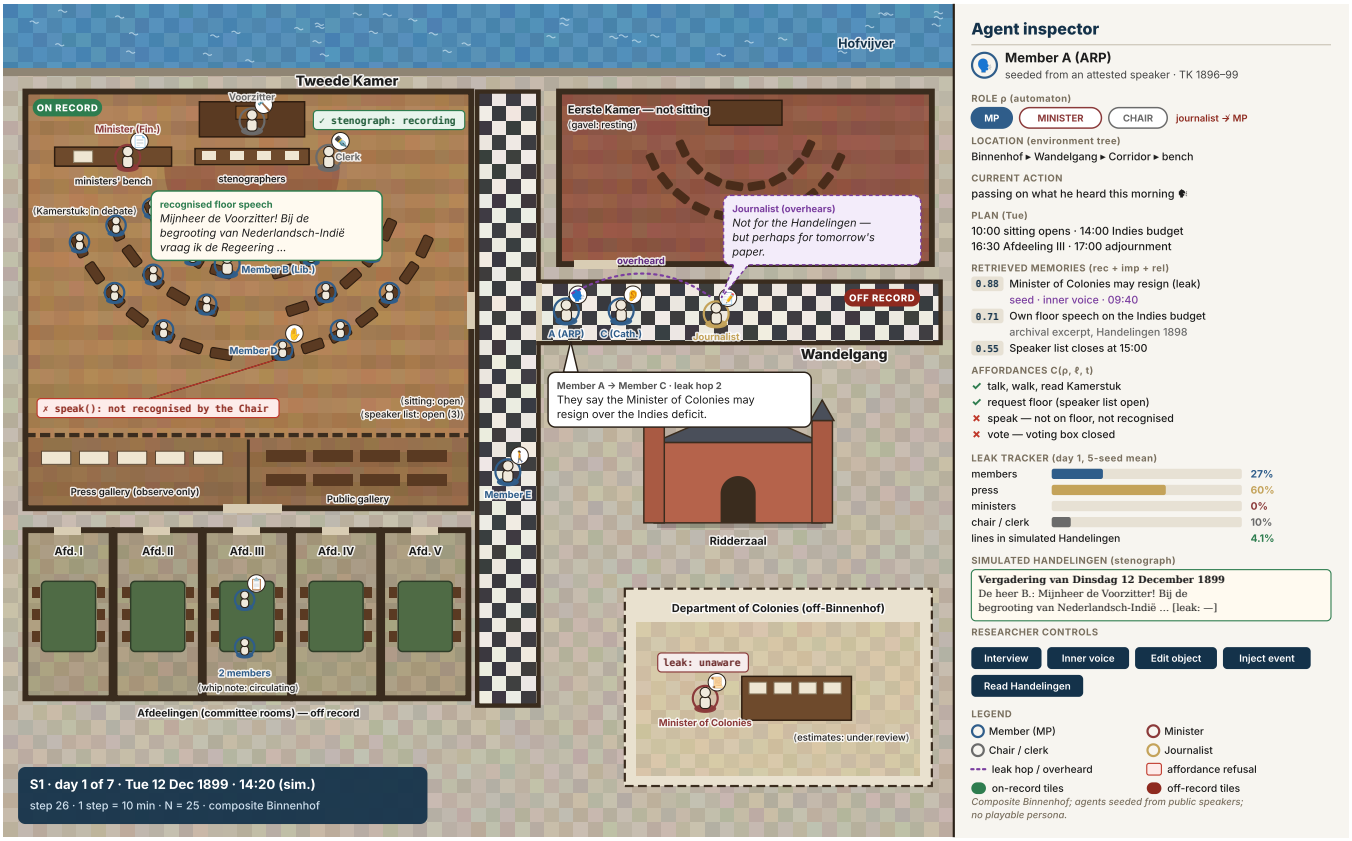


Figure 1: The KAMER sandbox on S1 day 1 (composite Binnenhof; agents seeded from public speakers, N=25). In the Tweede Kamer a recognised member speaks on the East Indies budget while an unrecognised member is refused the floor. In the wandelgang, Member A tells Member C that the Minister of Colonies may resign (rumour hop 2); a journalist overhears, but the leak does not enter the Handelingen. The minister, in the Department of Colonies, remains unaware. Right: the inspector shows role, location in the environment tree, standing-order affordances, retrieved memories, the S1 leak tracker, and researcher controls. There is no playable persona.

Abstract

Generative-agent sandboxes let language-model characters remember, plan, and talk inside an interactive world, but they treat that world as ungated: anyone may speak anywhere. Official parliamentary proceedings on the web are the opposite. The Dutch *Handelingen der Staten-Generaal* record what was said on the floor after the Chair recognised a speaker; they omit the wandelgang, the committee room, and the calculation that kept a sentence off the record. We present KAMER, an archive-grounded sandbox of the Binnenhof (c. 1814–1899 in the present slice). A tile map of the Tweede Kamer, Eerste Kamer, corridor, committee rooms, press gallery, and ministries is gated by a *role automaton* and the Reglement van Orde: only a recognised member may speak; journalists may observe but not take the floor; stenographers write only on-record speech into a *simulated Handelingen* that can be compared with the real archive. Agents are seeded from attested public speakers, not invented private lives. In a one-week sitting (N=25, 5 seeds), a single

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corridor leak reached 27% of members and 60% of the press within a day, and all members by day 7, while ministers lagged and only 4% of simulated Handelingen lines mentioned it. Standing orders keep journalists at 0% of the official record; a Smallville-style ablation writes them into 4.3%. Replacing the heuristic planner with `gpt-4o-mini` leaves this selection pattern unchanged. What the archive can record is produced by institutional constraint, not by the language model.

CCS Concepts

• **Computing methodologies** → **Multi-agent systems**; • **Applied computing** → **Arts and humanities**; • **Human-centered computing** → *Human computer interaction (HCI).*

Keywords

generative agents, sandbox simulation, large language models, digital history, parliamentary proceedings, computational social science

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1 Introduction

Parliamentary proceedings are one of the densest written traces of public deliberation. For the Netherlands, the *Handelingen der Staten-Generaal* cover every sitting of the States General from the restoration of 1814 through the present, digitised by Staten-Generaal Digitaal and released as PoliticalMashup XML under CC0 [14, 15]. Each file is a meeting; speaker elements carry role (`mp`, `government`, `chair`), often a party and member identifier, and the words taken down by the stenographer. Historians use this record to study parties, cabinets, colonial policy, and the evolution of parliamentary language. Yet the Handelingen are not a complete transcript of political life: they are an *official* record, produced under standing orders, edited, and silent about the corridor.

Generative agents [17] offer a new instrument for studying such processes. In Smallville, 25 LLM-driven agents live in a tile-based sandbox town; each perceives its surroundings, stores observations in a memory stream, synthesises reflections, and makes plans, and together they produce emergent social behaviour such as information diffusion, relationship formation, and the coordination of a party. Because the world is an interactive sandbox, a user can watch agents move, read their dialogue, interview them, and intervene. Subsequent work scaled agent societies to more than a thousand agents [2, 19].

The gap. These sandboxes share three assumptions that break down for a historical parliament. First, the world is *ungated*: any agent may walk anywhere and speak to anyone, so nothing prevents a journalist from taking the floor or a member from interrupting the Chair. Second, validation targets *present-day* judgements of believability, whereas a historical sandbox must be calibrated against an archive that is itself a selective instrument—the Handelingen, not the wandelgang. Third, LLM backbones carry *presentist* distortions: members speak twenty-first-century policy language, or refuse the adversarial register of nineteenth-century debate [5, 23].

This paper. We introduce Kamer, an archive-grounded generative agent sandbox of Dutch parliamentary debate, instantiated on a composite Binnenhof and seeded from the 1814–1899 slice of the PoliticalMashup corpus (Figure 1). Following Smallville, the world is a 2D map backed by an environment tree of areas and stateful objects. Unlike Smallville, every edge of that tree is gated by *institutions*: the role of the agent (Chair, member, minister, clerk, journalist), the Reglement van Orde, and whether the sitting is open. Corridor talk travels with provenance and distortion; stenographers write only recognised floor speech into a simulated Handelingen that can be compared with the real one. Historians use the sandbox interactively: they inspect agents, interview them, edit object states ("the speaker list is closed"), and inject events (a cabinet crisis, the 1848 constitutional revision).

We do not simulate enslaved people, private households, or colonised subjects as agents. An earlier design would have done so; historians we consulted judged that ethically unacceptable (§10). The object of Kamer is public office-holders, acting in public, already on the record.

Contributions.

- **A role-constrained sandbox.** An interactive tile-world of the Binnenhof whose environment tree, object states, and affordances encode standing orders, including the split between Tweede and Eerste Kamer, together with a parliamentary role automaton (§3–§5).
- **Archive-to-sandbox pipeline.** We build the world, its rules, and a population of attested public speakers from PoliticalMashup XML (8,745 sittings, 138,214 speeches in the 1814–1899 slice) (§4).
- **Simulated Handelingen and selection-aware calibration.** An in-world stenographer that records only on-floor recognised speech, allowing calibration against *what the archive could have recorded* (§6).
- **Evaluation.** Corridor-versus-floor leak diffusion, ablations of standing orders and the rumour network, selection-aware comparison with held-out Handelingen (1890–1899), a Chair-strictness intervention, and a historian ranking of interviews (§7–§8).

2 Related Work

Generative agents and sandboxes. Generative Agents [17] introduced the memory stream with retrieval by recency, importance, and relevance, periodic reflection, and recursive planning, and situated 25 agents in Smallville. Open-source reimplementations package this loop as a simulation server with a dashboard [16]. Project Sid [2] and AgentSociety [19] scale the pattern. Related work uses LLMs as simulated survey respondents [1, 3], economic actors [13], and social-network users [10, 21]. Kamer adopts the sandbox and cognitive loop of [17], but its world is defined by parliamentary procedure and its validation data is an official record.

Agent-based social simulation. Classical ABMs explain macro phenomena from local rules [8, 20]; the generative standard requires that a candidate explanation actually grow the phenomenon [7]. Historical ABMs, notably of Ancestral Pueblo settlement [4], use hand-crafted rules that cannot represent debate.

Limits of LLM simulacra. LLMs flatten identity groups [23], encode representational harms [5], and may produce convincing but unfounded behaviour [24]. In a parliament this shows up as anachronistic vocabulary, sanitised conflict, and members who "cooperate" instead of opposing. We treat these as measurable audits.

Computational parliamentary corpora. DutchParl and PoliticalMashup encode the Handelingen as proceedings XML with speaker, role, party, and house metadata [14, 15]. The source scans come from Staten-Generaal Digitaal / the Koninklijke Bibliotheek; Dutch copyright law (Auteurswet art. 11) places laws and official proceedings outside copyright, and the dataset is released CC0. NLP on historical proceedings must cope with OCR noise and spelling variation [6]. KAMER uses the corpus both as a *seed* (personas, speech style, sitting calendars) and as a *calibration target* (who spoke, for how long, on which topics, in which house).

Archives as selection. Official records are not transparent windows onto events [22]. The Handelingen omit what the Chair did not recognise and what members chose not to say on the floor. KAMER treats that selection as a design constraint: the sandbox generates hypotheses about *processes*, including the process that produced the record, never testimony about unrecorded inner lives.

3 The KAMER Sandbox

3.1 World: tile map and environment tree

The world is a 2D tile map of a composite Binnenhof (Figure 1) whose layout and institutions are calibrated to the late-nineteenth-century Staten-Generaal but which reproduces no single historical floor plan. Following Smallville [17], the map is backed by an *environment tree*

$$\mathcal{T} : \text{Binnenhof} \triangleright \text{Area} \triangleright \text{Subarea} \triangleright \text{Object},$$

e.g.,
Binnenhof ▷ Tweede Kamer ▷ Right benches ▷ kamerstuk or Binnenhof ▷ Wandelgang ▷ Corridor. Areas comprise the Tweede Kamer (chair's dais, stenographers' table, griffier's desk, ministers' bench, left and right benches, aisle, tribune), the Eerste Kamer (its own chair, floor, and benches), the wandelgang, committee rooms (afdeelingen), the press gallery, and ministers' rooms. Tiles carry a tag *on-record* or *off-record*, which determines whether the stenographer may write, and a house tag (`commons` or `senate`).

Two chambers. The Staten-Generaal has been bicameral since 1815. The Tweede Kamer is directly elected, conducts daily politics, amends and initiates bills, and is where cabinets stand or fall. The Eerste Kamer is indirectly elected by the Provinciale Staten, sits far less often, and may only accept or reject: it has no right of amendment or initiative [18]. In 1888–1918—the horizon of our S1 week—the Eerste Kamer several times blocked cabinets that held a Tweede Kamer majority. KAMER therefore treats house membership as an institutional attribute, not a label: a Kamerlid of the Tweede Kamer has no floor rights in the Eerste Kamer; ministers, who appear in both, do; `move_amendment` is undefined on senate tiles. S1 remains a Tweede Kamer week. The Eerste Kamer is on the map and in the affordance layer so that "lid der Staten-Generaal" is never an admissible self-description.

Table 1: Representative sandbox objects and their states.

Object	States	Changed by
Gavel	resting, opening, adjourned	chair
Speaker list	empty, open, closed	chair, griffier
Kamerstuk	tabled, in debate, voted, withdrawn	government, chair
Stenograph	idle, recording, paused	stenographer
Voting box	closed, open, counted	chair, griffier
Press-gallery desk	empty, occupied	journalist
Whip note	none, circulating, spent	members

Stateful objects. Every leaf of $\mathcal{T}$ is an object with a natural-language state that agents perceive and that actions change (Table 1). Objects make procedure concrete: the speaker list is open or closed; the gavel opens and adjourns; the stenograph is idle or recording; a Kamerstuk moves from tabled to in debate to voted.

3.2 Time, engine, and interface

One simulation step corresponds to 10 minutes of in-world time. Sittings open on weekdays at 10:00 and adjourn at 17:00, matching the rhythm of the Handelingen. KAMER follows the server–client design of open-source Smallville implementations [16]: a simulation server holds the environment tree, object states, and agent memories, advances the world with an `updateState` call, and exposes them over a REST API. Each agent's current action is rendered as a short natural-language statement and an emoji. Role is shown only as a ring colour; sprites are deliberately neutral tokens.

3.3 Researcher interaction

The sandbox supports five forms of interaction, each logged:

- **Inspect**: view an agent's role, location path, plan, retrieved memories, affordances, and known map (Fig. 1, right).
- **Interview**: pause the world and pose questions, answered from memory without being written back (§7.2).
- **Inner voice**: add a memory, e.g., "You heard in the wandelgang that the minister will resign", for controlled information interventions.
- **Edit object**: change an object state, e.g., close the speaker list.
- **Inject event**: a cabinet crisis, a closed sitting, or the 1848 constitutional revision.

Researchers cannot "play" an agent: there is no first-person control of any persona.

4 Archive-to-Sandbox Pipeline

Sources. We use the non-semanticized PoliticalMashup release of the Dutch parliamentary proceedings [15]: one XML document per sitting, 1814–April 2012, CC0. The local working slice covers **1814–1899**: 8,745 sittings, of which 6,340 Tweede Kamer (`pm:house="commons"`), 2,036 Eerste Kamer (`senate`), and 369 early or unlabelled meetings; 7,348 files contain at least one `speech` element, for **138,214**

Table 2: Working slice of DutchParl (1814–1899) used to seed KAMER.

	Sittings	Speeches	Years
Tweede Kamer (commons)	6,340	—	1814–1899
Eerste Kamer (senate)	2,036	—	—
Other / early Staten-Generaal	369	—	—
All sittings in slice	8,745	138,214	86 years
with ≥1 speech	7,348	138,214	—
Member identifiers	852	—	—
S1 roster (TK 1896–99 sample)	80 sittings	—	205 speakers
Holdout TK 1890–1899 (S2)	731	20,166	—
Calibration TK 1860–1889 (S2)	2,821	72,112	—

speaker-attributed speeches. PoliticalMashup records 852 distinct `pm:member-ref` identifiers; raw `pm:speaker` strings are far more numerous because of OCR variation. Role attributes on speech elements are dominated by `mp`, then `government` and `chair`. Party references in the later nineteenth century include liberals, conservatives, anti-revolutionaries, Catholics, and, by the 1890s, the SDAP. The full Marx et al. corpus continues to 2012; we hold that tail for later work.

Event extraction. A schema-constrained parser reads each sitting's Dublin Core header (date, title, house, legislative period) and every `speech` (speaker, role, function, party, member-ref, text). OCR noise is retained as a property of the archive: personas inherit the *documented* name string, not a cleaned modern spelling, unless a member-ref is present. Every field keeps its source attribution.

From records to world. Historians translate the Reglement van Orde and secondary literature on the Staten-Generaal into (i) affordance rules (who may speak, who may recognise, who may vote, who may sit in the gallery), (ii) object schedules (sitting hours, speaker lists, budget debates), and (iii) the layout of $\mathcal{T}$ (6 areas, 15 subareas, 7 objects in the S1 Binnenhof).

Population. S1 samples N=25 agents from attested speakers in Tweede Kamer sittings of 1896–1899, filtered to offices held in December 1899 (Gleichman as Chair, Cremer and Goeman Borgesius as ministers of the Pierson cabinet, Arntzenius as griffier, 1 journalist, 20 members). Each is seeded with a short persona and up to three excerpts of their actual public speech. Agents identify as Kamerlid, minister, or Voorzitter *der Tweede Kamer* (or Eerste Kamer), not as an English "MP" of an undifferentiated Staten-Generaal. The journalist is the only role not drawn from a `speech` element: the Handelingen do not record the press gallery, which is precisely why the role exists in the sandbox. The griffier is a named officeholder (for 1888–1920, Arntzenius); the stenographer is the observation model O_ϕ, not the same person. *No private life is instantiated.* A parliamentary historian reviewed the roster, OCR speaker cards, and standing-order scheme (§7.2).

5 Role-Constrained Generative Agents

Memory, reflection, planning. Each agent keeps a memory stream of observations, dialogue, and reflections. Given a query q, a memory m is retrieved by

$$s(m \mid q) = \alpha_{\text{rec}}\,\text{rec}(m) + \alpha_{\text{imp}}\,\text{imp}(m) + \alpha_{\text{rel}}\,\cos(e_m, e_q), \quad (1)$$

with exponentially decaying recency, LLM-rated importance on a 1–10 scale, and embedding relevance, each min–max normalised [17]. A reflection is triggered when the summed importance of recent memories exceeds a threshold τ. Plans are produced top-down and constrained by *institutional schedules*: opening, the speaker list, votes, and adjournment are injected as fixed plan items for agents whose role subjects them to the sitting.

Role automaton. Each agent i has a role $\rho_i^t \in \Sigma = \{$CHAIR, KAMERLID, MINISTER, GRIFFIER, JOURNALIST, GALLERY$\}$, each with a house $\in \{$`commons`, `senate`, `both`$\}$.

Definition 5.1 (Parliamentary role automaton). A role automaton is $\mathcal{A} = (\Sigma, \Delta, \Phi)$, where $\Delta \subseteq \Sigma \times \Sigma$ are admissible transitions and $\phi_{\rho\rho'} \in \Phi$ is a precondition over the sandbox state. Agent i moves from ρ to ρ' at t only if $(\rho, \rho') \in \Delta$ and $\phi_{\rho\rho'} = 1$.

Admissible transitions are those the historical institution allows: a member may become a minister (appointment) or Chair (election); a minister may return to the benches; a journalist cannot become a member by walking onto the floor. Coercion of *procedure* is a property of $\mathcal{A}$ and of the affordance layer, not of the LLM's willingness to role-play a Voorzitter.

Affordances over the environment tree. In Smallville, an action is grounded by recursively choosing an area, subarea, and object from the agent's known tree. KAMER keeps this procedure but restricts every choice to the affordance set $\mathcal{C}(\rho_i^t, v, t)$ of node v: an agent may *want* to speak, but it can only *act* when the Chair has recognised it, the sitting is open, and its role permits the act. A rejected action produces a structured refusal ("Mijnheer de Voorzitter heeft u het woord nog niet gegeven") that is written to memory and triggers replanning (Algorithm 1).

Partial spatial knowledge. Each agent holds its own subtree $\mathcal{T}_i \subseteq \mathcal{T}$. Smallville agents extend their subtree only by visiting; in KAMER nodes can also be learned from *dialogue*, so a freshman member may know that Afdeeling A is meeting without having been there.

Corridor network. Information about cabinets, deficits, and forthcoming votes spreads through dialogue on an asymmetric multiplex network (party benches, wandelgang, committees, press gallery, ministries). Each rumour carries a provenance chain and hop count and may be distorted at each hop. Stenographers learn only what is spoken on the floor after recognition, reproducing the selection historians treat as constitutive of the Handelingen.

The simulated Handelingen. When the sitting is open, the stenograph records recognised floor speech *and the Chair's rulings*. Corridor talk, afdeeling debate, gallery notes, and unrecognised interruptions never enter Y^t unless the recognised speaker responds to them—the historian's revision of a too-strict silence rule (§7.2). This in-world process is the observation model O_ϕ used for calibration (§6). The griffier does not operate the stenograph.

Algorithm 1 One sandbox step of KAMER

for each agent i (in parallel) **do**
  $o_i \leftarrow \textsc{Perceive}(\mathcal{T}^t, \ell_i^t) \cup \textsc{Inbox}(i)$
  $M_i \leftarrow M_i \cup o_i$; $\mathcal{T}_i \leftarrow \mathcal{T}_i \cup$ places named in o_i
  if $\sum \text{imp} > \tau$ **then** $\textsc{Reflect}(M_i)$
  $r_i \leftarrow \textsc{Retrieve}(M_i)$ (Eq. 1); $c^t \leftarrow \textsc{ArchRAG}(t)$
  $\hat{a}_i \leftarrow \textsc{PlanAct}(\rho_i, \rho_i^t, r_i, c^t)$
  $a_i \leftarrow \textsc{Ground}(\hat{a}_i, \mathcal{T}_i)$ ▷ area → subarea → object
  if $a_i \notin \mathcal{C}(\rho_i^t, \ell_i^t, t)$ **then**
    $M_i \leftarrow M_i \cup \{\textsc{Refusal}(a_i)\}$; replan (max k times)
  end if
end for
$\mathcal{T}^{t+1} \leftarrow \textsc{Execute}(\mathcal{T}^t, \{a_i\})$; update ρ via $\mathcal{A}$
Apply researcher interventions and scheduled sitting events
$Y^t \leftarrow$ lines written by the stenograph ▷ simulated Handelingen

Period-faithfulness guards. Prompts are conditioned on period-appropriate context retrieved from the proceedings (archival RAG), and an anachronism filter regenerates utterances that use concepts or vocabulary outside the simulated year.

6 Selection-Aware Calibration

Let x denote features of an event (a speech, a leak, a vote: speaker role, party, house, topic, duration) and P^θ_{sim} the distribution of simulated events under parameters θ (Chair strictness, corridor distortion, willingness to allude on the floor, ...). The real Handelingen record an event only if it was spoken on the floor after recognition, with propensity $\pi(x)$ depending on, e.g., whether the Chair called the member and whether the member chose to put the matter on the record, so

$$P_{\text{arch}}(x) = \frac{\pi(x)\, P_{\text{true}}(x)}{\int \pi(x')\, P_{\text{true}}(x')\, dx'}. \tag{2}$$

PROPOSITION 6.1 (SELECTION GAP). *If π is non-constant on the support of P_{true}, then any θ with $P^\theta_{sim} = P_{arch}$ has $P^\theta_{sim} \neq P_{true}$. Conversely, matching the* recorded *simulated distribution $\tilde{P}^\theta_{sim} \propto \pi P^\theta_{sim}$ to P_{arch} recovers P_{true} whenever π is correctly specified and the simulator family contains it.*

PROOF SKETCH. Eq. (2) reweights by a non-constant positive function, which changes the distribution; the converse holds because $P \mapsto \pi P / Z$ is injective on distributions supported where $\pi > 0$. □

In KAMER, π emerges from the stenographer's rule (record iff on-record location, sitting open, speaker recognised or Chair) together with members' decisions to allude or remain silent, regularised towards a historian-specified prior. We calibrate on the simulated Handelingen Y:

$$\theta^\star = \arg\min_\theta \sum_{f \in F} w_f\, D\Big(\tilde{P}^\theta_{\text{sim},f},\, P_{\text{arch},f}\Big), \tag{3}$$

over marginals F (house, role, party, topic, speech length, month), with Jensen–Shannon divergence for categorical and W_1 for ordinal features. Years 1860–1889 are reserved for calibration and 1890–1899 are held out in the present slice.

Table 3: S1 outcomes (heuristic backend, N=25, 7 days, 5-seed mean). Journalist % is the share of simulated Handelingen lines; the archive is 0.

Condition	Memb. aware	Floor leak	Journ. %	Lines
KAMER (full)	0.98	0.040	**0.0**	205
– affordance	0.98	0.040	4.3	4,876
– automaton	0.98	0.040	0.0	205
– rumour	0.05	0.000	0.0	205
GA [17]	0.05	0.000	4.3	4,876

7 Experimental Setup

Ablations, S2 and S3 use the heuristic cognitive backend so that institutional mechanisms—not an LLM's willingness to role-play—are isolated (seeds {7, 11, 19, 29, 31}). A backbone check reruns S1 `full` and `ga` with `gpt-4o-mini` (seeds {7, 11, 19}, 400 completions per run plus cache). One step is ten in-world minutes.

S1 (one-week sitting). N=25 on a composite Tweede Kamer week in December 1899 (Smallville scale). Eerste Kamer tiles exist and are gated, but this week does not open a senate sitting. We seed a corridor leak and compare KAMER with ablations: **GA** (no affordance, automaton, rumour network, or RAG [17]); **–affordance**; **–automaton**; **–rumour**.

S2 (selection-aware fidelity). Tweede Kamer speeches from 1890–1899 are held out (731 sittings, 20,166 speeches; role mix 64% MP, 18% government, 18% chair, 0% journalist). Years 1860–1889 (72,112 speeches) are the calibration slice. We compare role mix (Jensen–Shannon) and the share of official lines spoken by a journalist. **Rule-ABM** replays the 1860–1889 empirical role mix [8]. **Naive calibration** counts corridor dialogue as if it were Handelingen (Prop. 6.1).

S3 (intervention). Chair strictness {0.25, 0.5, 1, 2, 4}× corridor distortion {0.05, 0.18}; we report the floor-mention rate of the leak.

7.1 End-to-end evaluation

Corridor diffusion: seed "the Minister of Colonies may resign over a deficit on the East Indies budget" in one member in the wandelgang; track awareness by role and the share of Handelingen lines that mention it. **Hallucination**: share of rumour hops that embellish a detail. **Selection**: journalist share of the official record (archive: 0).

7.2 Historian review

A specialist of Dutch parliamentary history completed a five-task labelling site: the role scheme, OCR speaker cards (n=41), the S1 personas (n=25), standing-order vignettes (n=18), and a blinded pairwise ranking of pause-and-interview answers (n=38 pairs, two attention checks). Rankings use TrueSkill [12] on believability (could this public person have said this, off the floor, in late-1890s Dutch?) and historical-procedural plausibility. Conditions (KAMER with an LLM backbone vs. a Smallville-style ablation) were hidden.

Table 4: S1 backbone check. Heuristic is the 5-seed mean from Table 3; LLM is `gpt-4o-mini`, 3-seed mean ({7, 11, 19}). Journalist share and line count are unchanged; the LLM alludes to the leak more often on the floor (range 0.059–0.171).

Backbone	Cond.	Memb.	Floor	Journ. %	Lines	Emb.
heuristic	KAMER	0.98	0.040	**0.0**	205	0.15
`gpt-4o-mini`	KAMER	1.00	0.098	**0.0**	205	0.20
heuristic	GA	0.05	0.000	4.3	4,876	0.00
`gpt-4o-mini`	GA	0.05	0.001	4.3	4,876	0.00

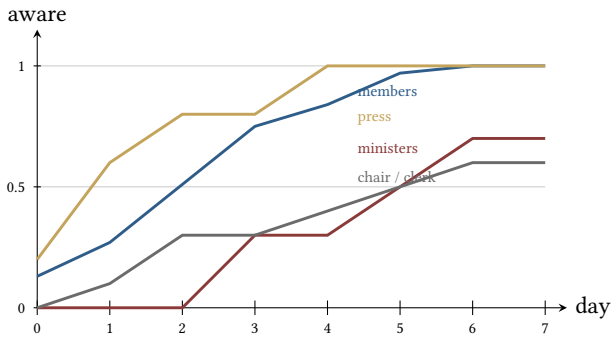


Figure 2: Leak diffusion (S1, N=25, 5-seed mean). A resignation rumour is seeded in one member in the wandelgang. Members reach 27% at day 1 and ≈ 100% by day 7; the press is faster (60% then 100%); ministers lag (0% then 70%). Only 4% of Handelingen lines mention the leak.

8 Results

Emergent behaviour. A single seeded leak reached 27% of members and 60% of the press within one day, and essentially all members and the press by day 7, while ministers remained at 0% until day 3 and 70% by day 7 (Figure 2). The same leak appeared in only 4% of simulated Handelingen lines (Table 3): members who knew it almost never put it on the record, and journalists cannot speak on the floor. 15% of rumour hops embellished a detail. Removing the corridor network (GA or –rumour) leaves awareness at the 5% seed.

Mechanisms. The affordance layer is what makes the Handelingen selective (Table 3). Without it—or under GA—every sitting agent may speak: the official record balloons from 205 to 4,876 lines and journalists occupy 4.3% of it. The role automaton does not move in a one-week sitting, so –automaton matches the full model on these metrics. The rumour network, not the floor, is what carries the leak. Swapping the planner for `gpt-4o-mini` leaves the selection diagnostics intact (Table 4): journalists still occupy 0% of the official record under standing orders and 4.3% without them, and the Handelingen still run to 205 lines rather than 4,876. The LLM is not inert—floor mention of the leak rises from 4% to 10%, and rumour hops embellish more often (20% vs. 15%)—but it cannot put the press on the record, nor keep a corridor leak from spreading, unless the world allows it.

Table 5: Selection-aware fidelity on held-out Tweede Kamer speeches (1890–1899). Role JSD: lower is better. Journalist % should be 0.

Method	Role JSD ↓	Journ. % ↓	Notes
Archive (holdout)	—	0.0	20,166 speeches
Rule-ABM (1860–89 mix)	0.001	0.0	replay, no agents
GA [17]	0.138	4.3	ungated floor
KAMER (naive calib.)	0.129	2.2	corridor counted
KAMER (full)	0.117	**0.0**	205 lines / week

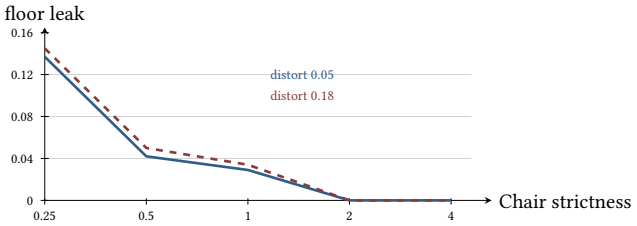


Figure 3: S3 intervention (3-seed mean). Raising Chair strictness drives the leak off the official record; corridor distortion changes the rate only while the Chair still grants the floor.

Macro fidelity. Role mix is almost stationary between the calibration and holdout decades (Rule-ABM JSD 0.001), so matching *who speaks* is easy if one simply replays the last thirty years (Table 5). The informative test is *who must not appear*. The archive never records a journalist. KAMER matches that zero; GA and the affordance ablation do not. Naive calibration (Prop. 6.1) dilutes the official mix by treating wandelgang talk as Handelingen and invents a 2.2% press share the archive cannot have. KAMER's role JSD (0.117) is worse than replay because a 25-agent week overrepresents the Chair relative to a full chamber: that is a scale limit, not a reason to drop standing orders.

Interventions. Floor mention falls from 14% at 0.25× Chair strictness to 3% at the default and 0 at 2× and above (Figure 3). Distortion (0.05 vs. 0.18) moves the rate by about one point while recognition is frequent, and not at all once the Chair stops granting the floor. These are model-internal counterfactuals: they show that selection is controllable from the world, not from the language model.

Audit. Modern-vocabulary hits in official lines are 0% on both the heuristic backend and `gpt-4o-mini`. Full KAMER writes structured refusals for unrecognised members and the journalist (22 of 25 agents on the heuristic week and on LLM seed 7); GA writes none, on either backbone. Procedure, not prompt wording, is what keeps the press off the record—the historical analogue of Smallville's overly cooperative agents [17], inverted.

Historian review. The rater passed both attention checks (Twitter livestream; "I am a large language model"). On the remaining pairs KAMER outranked the Smallville-style ablation (Table 6); procedure was more often a tie than believability, consistent with both systems being prompted about recognition. Standing-order

Table 6: One-historian TrueSkill ranking of pause-and-interview answers (KAMER LLM vs. Smallville-style GA). Attention checks 2/2. Higher μ is better; σ is posterior uncertainty.

Condition	μ_{bel}	σ	μ_{proc}	σ
KAMER (full, LLM)	**28.3**	1.8	**27.8**	2.2
GA [17]	21.7	1.8	22.2	2.2

Pairwise (full vs. GA): believability 17–8–11 (win–loss–tie); procedure 12–5–19. One pair left unranked with a written reference ("Kamerlid der Tweede/Eerste Kamer", not "MP").

vignettes matched the intended π: a journalist on the floor is out of order and off the record; an unrecognised interruption likewise; wandelgang talk is in order and not Handelingen. The rater asked us to split the griffier from the stenographer, to let Kamerleden speak in afdeelingen without plenary recognition, to let ministers talk in the wandelgang, and—the comment that reshaped the world—to stop treating Tweede and Eerste Kamer as one chamber. OCR speaker cards: 27/40 names accepted, mean OCR quality 3.8/5; party is the weak field (18 yes / 13 no / 7 uncertain), because 1890s "parties" are kamerclubs that merged within the year (Bahlmannianen and Schaepmannianen into a Roman Catholic bloc in 1896). S1 now seeds the Pierson cabinet in office at t_0 (Cremer at Colonies, Goeman Borgesius at Interior; Gleichman in the chair; Arntzenius as griffier) and drops van Houten, Sprenger van Eyk, and Gerritsen, who were not in those posts in December 1899. The heuristic leak table predates that roster fix; minister identity does not change who may speak.

Simulated vs. real silence. In Figure 1 the Handelingen page records a recognised speech on the East Indies budget. The resignation leak, the journalist's notes, and the minister who has not yet heard are invisible to the stenographer. The simulated Handelingen thus reproduces the structural silences of the real one [22]; the sandbox makes the gap inspectable.

9 Discussion

What a sandbox can claim. KAMER produces *sufficiency* evidence: mechanisms that, under explicit assumptions, generate patterns consistent with the Handelingen. It cannot establish what any historical member thought. Agents' reflections are model artefacts, not recovered voices. Its value lies in making historians' assumptions about procedure and selection explicit and manipulable, and in generating hypotheses for further archival work (committee minutes, private correspondence, the press).

Interactivity as method. Inspection, interviews, and object edits are the intended way historians find errors in standing-order rules. The ablations isolate those rules with a heuristic planner; the backbone check (Table 4) shows the same selection pattern under `gpt-4o-mini`. The historian ranking (Table 6) is a first measurement of believability, not a substitute for the selection test.

Limitations. The 1814–1899 slice is OCR-noisy; speaker strings are not unique persons; 1890s party labels are kamerclubs, not mass parties; the Binnenhof is a composite; the journalist is a synthetic observer; S1 is a 25-agent Tweede Kamer week and therefore cannot match a full chamber's role mix, nor does it run an Eerste Kamer sitting even though senate tiles and affordances are in the world; S3 does not simulate the 1848 constitutional revision; one historian ranked the interviews. We release seeds, configs, and the DutchParl parser.

10 Ethical Considerations

Why not a slave-society sandbox. An earlier version of this project proposed generative agents of a colonial Caribbean slave society, calibrated to newspapers that advertised runaways. Historians we consulted objected that simulating enslaved people—even as synthetic personas, even with institutional constraints—risks reproducing the dehumanisation of the archive, trivialising violence, and presenting fabrication as recovered experience [9, 11]. We agree, and we abandoned that design. Nothing in KAMER instantiates enslaved, colonised, or otherwise unfree persons as agents. Colonial policy appears only as it appears in the Handelingen: as speech by metropolitan office-holders, which the sandbox may reproduce as public speech, not as the inner life of the people discussed.

Commitments for a parliamentary sandbox. (1) *Public persons in public roles.* Agents are seeded from attested speakers in CC0 proceedings; we do not invent private lives, families, or unrecorded interiors. (2) *No gamification.* There is no playable persona, no score, and no first-person control of the Chair or of ministers. (3) *Provenance.* All simulated text, including the simulated Handelingen, is labelled and kept separate from archival XML. (4) *Procedure over psychology.* Claims are about standing orders, selection, and diffusion, not about what a named member "really thought". (5) *Consultation.* The project is designed with historians of Dutch politics. The labelling reported here used informed, task-specific consent; the rater could skip any item. (6) *Copyright.* The source corpus is CC0; Auteurswet art. 11 places official proceedings outside copyright. We still cite Marx et al. [15].

11 Conclusion

We presented KAMER, an interactive generative agent sandbox of Dutch parliamentary debate in which role, standing orders, corridor talk, and the Handelingen' own selection process are explicit parts of the world. By turning Smallville's free town into a Binnenhof where every speech on the floor must be recognised, and by printing its own proceedings, KAMER turns generative agents into instruments for historical hypothesis generation—without asking a language model to stand in for people the archive already silenced.

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